

Turn-On Metal Ion Responsive Behavior Enabled by Coordination-Driven Multicomponent Self-Assembly

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Shaozhi Wang, Manman Dai, Ning Zhang, Ziteng Guo, Desheng Li, Jingsong Yuan, Yibin Sun, Ningxu Han,* and Ming Wang*



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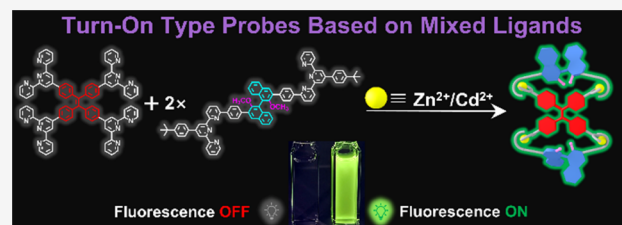


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ABSTRACT: The advancement of industrialization, fossil fuel reliance, and agricultural expansion has intensified environmental contamination by heavy metal ions, and they demand urgent attention. In this work, we proposed a multicomponent-assembly-based strategy for the construction of a turn-on fluorescence sensing system and designed mixed-ligand probes with tetraphenylethylene (TPE) and binaphthyl units. As many as 17 different metal ions were introduced into the mixed ligands; only Zn(II) and Cd(II) induced a significant fluorescence enhancement. The experimental results indicated that the binaphthyl-induced spatial restriction after coordination markedly enhanced the emission efficiency. Heteroleptic assemblies **S2** and **S4** ($\Phi_F = 65.7\%$ for **S2** and $\Phi_F = 49.8\%$ for **S4**) based on mixed ligands **L2** and **L3** displayed a 100-fold enhancement in fluorescence efficiency relative to the free ligand **L2** ($\Phi_F = 0.5\%$). Furthermore, according to fluorescence titration experiments, the calculated limits of detection (LODs) for Zn(II) and Cd(II) of our mixed (**L2** and **L3**) ligand sensing system were approximately 1.7×10^{-7} and 1.8×10^{-7} M, which are markedly superior to the **L1** and **L3** ligand combination. Consequently, mixed ligands **L2** and **L3** can serve as ideal turn-on fluorescent probes for the detection of Zn(II) and Cd(II) ions in a fast, sensitive, and selective manner.



INTRODUCTION

The rapid advancement of industrialization, continued reliance on fossil fuel, and expansion of agricultural activities have markedly intensified environmental contamination by heavy metal ions, demanding urgent attention.^{1,2} Many heavy metal ions are highly toxic and pose substantial risks to ecosystems and human health.³ Cadmium (Cd), for example, is harmful at low daily intake levels ($60\text{--}70 \mu\text{g}\cdot\text{kg}^{-1}$).⁴ Exceeding permissible limits can lead to chronic poisoning, organ dysfunction, and conditions such as Itai-itai disease, a notorious example of Cd poisoning characterized by severe bone pain and deformation.^{5–8} Therefore, it is essential to detect trace heavy metal ions quickly, flexibly, and accurately to prevent their excessive release and environmental accumulation.

"Turn-on" fluorescence sensing is widely regarded as an ideal detection strategy owing to its unique signal response mode, in which fluorescence switches from an "off" state (low or negligible luminescence) to an "on" state (manifestly bright luminescence) upon analyte recognition,^{9–13} enabling high sensitivity with minimal background interference. However, conventional dye molecules inherently exhibit bright emission in the dilute state and suffer from aggregation-caused quenching (ACQ) at high concentrations,^{14–18} which

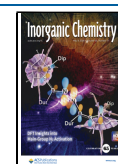
fundamentally conflicts with the operational principle of turn-on sensing. Aggregation-induced emission (AIE) materials offer an alternative paradigm^{19–23} as their fluorescence enhancement originates from the restriction of intramolecular motion in the aggregated state, which suppresses nonradiative decay pathways.^{24–28} Typically, a pronounced AIE behavior emerges only at sufficiently high concentrations, where molecular aggregation becomes favorable. This intrinsic concentration dependence, however, presents a fundamental limitation for chemical sensing, which usually requires operation under dilute conditions to detect trace analytes. Under such circumstances, aggregation is thermodynamically unfavorable, making conventional AIE-based turn-on strategies ineffective. Consequently, a key challenge in AIE-based sensing is how to impose effective restriction of intramolecular motion on AIE luminogens in dilute solution without relying on

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Scheme 1. Structures of Ligands and Self-Assembly of Complexes

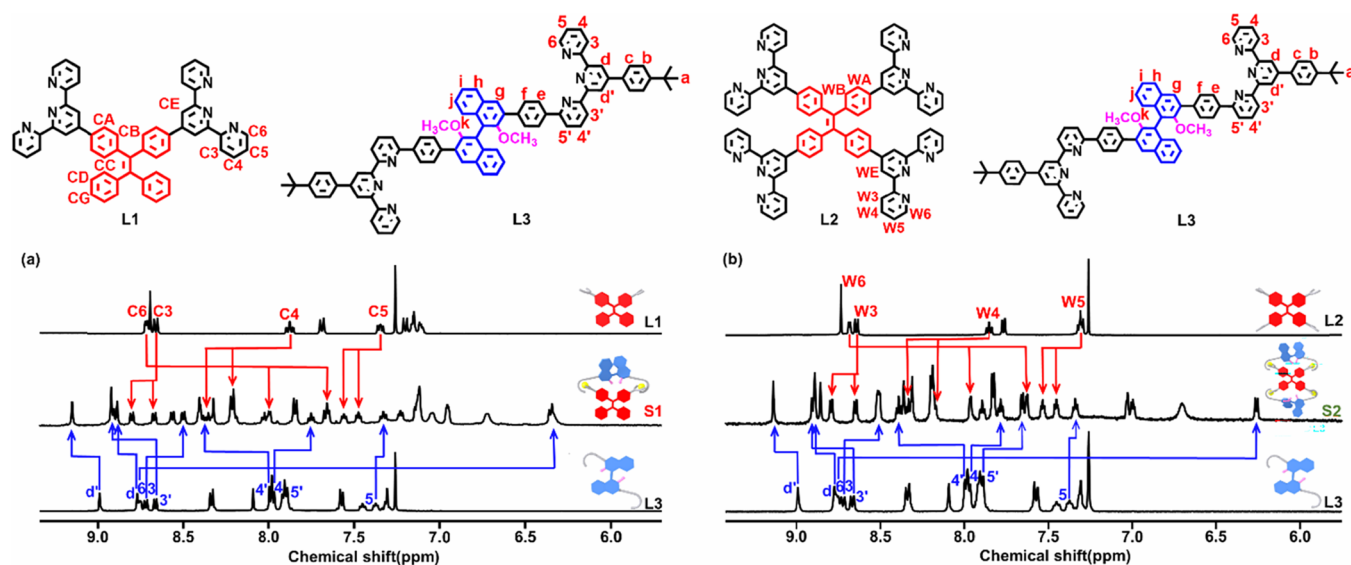
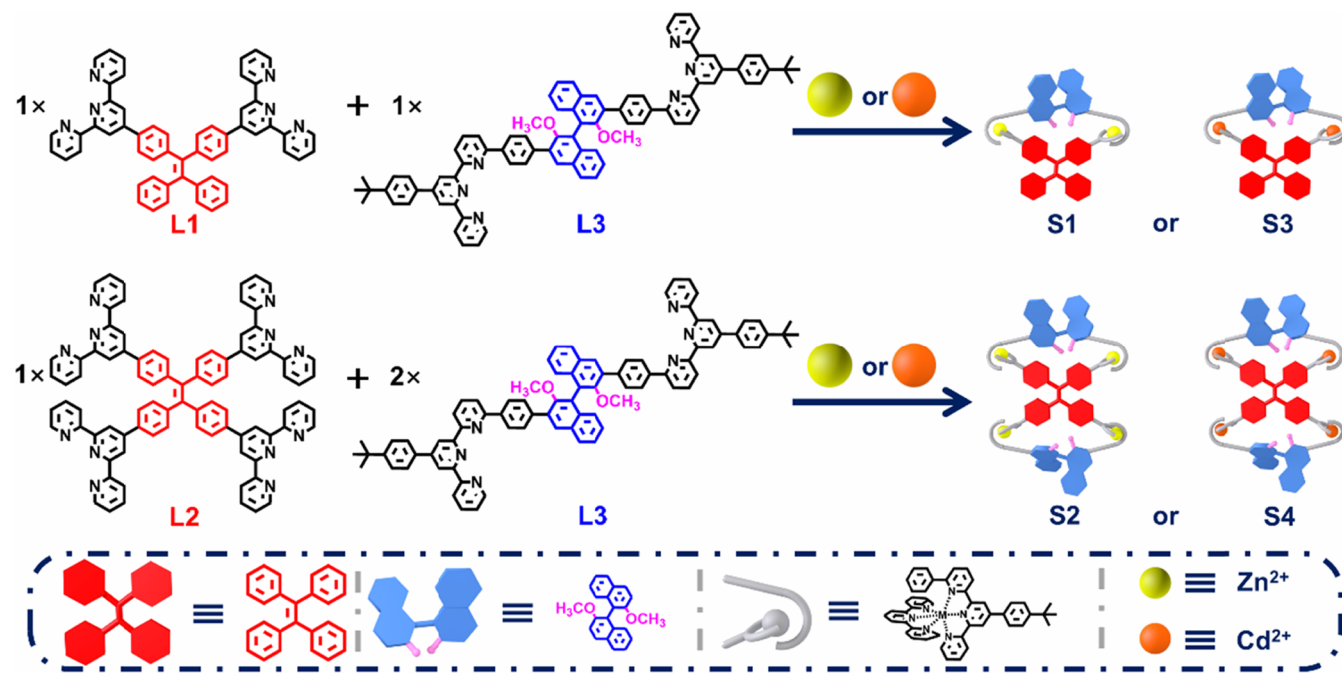


Figure 1. (a) ^1H NMR spectra (500 MHz, 300 K) of ligands **L1** and **L3** in CDCl_3 and complex **S1** in CD_3CN . (b) ^1H NMR spectra (500 MHz, 300 K) of ligands **L2** and **L3** in CDCl_3 and **S2** in CD_3CN .

aggregation. Addressing this challenge calls for the construction of a confined microenvironment that mimics the motion-restricted state of aggregates while maintaining molecular discreteness, thereby enabling turn-on fluorescence at low concentrations.

In this study, we proposed and designed a terpyridine (tpy)-based heteroleptic supramolecular assembly with efficient luminescence to achieve turn-on fluorescence sensing. The tetraphenylethylene (TPE) was incorporated as the core fluorescent moiety and engineered into ditopic ligand **L1** and tetratopic ligand **L2**. By employing a shape-complementary self-assembly strategy, the racemic ditopic binaphthyl-based²⁹ ligand **L3** can assemble with ligands **L1/L2** and Zn(II)/Cd(II) to form heteroleptic supramolecular complexes Zn(II)-based **S1** and **S2** and Cd(II)-based **S3** and **S4** (Scheme 1). Single-

crystal characterization of (S)-**S2** reveals that, upon coordination, the binaphthyl moiety can be precisely positioned adjacent to TPE in dilute solution. This configuration effectively restricts the intramolecular motion of TPE, leading to significant fluorescence enhancement. The photoluminescence quantum yield (PLQY) increases markedly from 0.5% for free ligand **L2** to 65.7% for **S2** and 49.8% for **S4**. In contrast, although **S1** and **S3**, which leave two TPE benzene rings unrestricted, still display significant fluorescence enhancement, the improvement in fluorescence quantum efficiency (15.0% for **S1** and 15.7% for **S3**) is significantly lower than that observed for **S2** and **S4**. Consequently, the combination of ligands **L2** and **L3** is used as a novel probe and exhibits superior detection performance toward Zn(II) and Cd(II) ions.

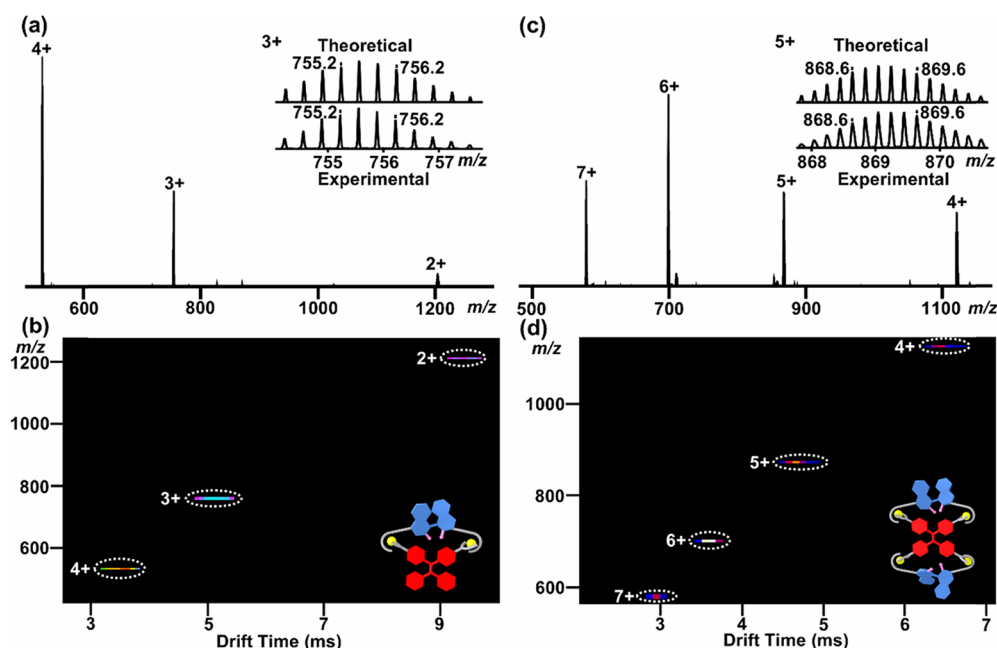


Figure 2. (a) ESI-MS spectrum of **S1** (inset: theoretical and experimental isotopic patterns for 3+ moiety). (b) TWIM-MS plot (m/z vs drift time) of **S1**. (c) ESI-MS spectrum of **S2** (inset: theoretical and experimental isotopic patterns for 5+ moiety) and (d) TWIM-MS plot (m/z vs drift time) of **S2**.

RESULTS AND DISCUSSION

We designed and synthesized TPE-based ditopic ligand **L1** and tetratopic ligand **L2**, as shown in Scheme 1. As a classic AIE fluorophore, TPE has fluorescence enhancement properties under spatial restriction,^{30–32} making it suitable as a core fluorescent component for turn-on sensors. Both ligands were obtained via an efficient Suzuki coupling reaction. To better achieve coordination-induced fluorescence enhancement and realize effective turn-on sensing, restricting the free motion of the TPE became crucial. Based on previous reports from our group, 6-position modification of tpy can improve coordination selectivity through a shape-complementary assembly strategy and enable the close packing of two core groups within different ligands.^{33–36} Accordingly, we designed and synthesized the racemic binaphthyl-based ligand **L3** based on the dissymmetrical modification of tpy. All ligands were synthesized in decent yields through the synthetic procedures in Scheme S1 and fully characterized by ^1H and ^{13}C NMR spectroscopy and matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry (Figures S2–S14). Complexes **S1** and **S2** were assembled by mixing ligand **L1** or **L2**, ligand **L3**, and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in a precise stoichiometric ratio ($\text{L1/L3/Zn}^{2+} = 1:1:2$ or $\text{L2/L3/Zn}^{2+} = 1:2:4$) in CHCl_3 and MeOH ($v/v = 1:2$) at 50°C for 10 h. After that, excessive NH_4PF_6 was added to the reaction solution to afford light-yellow precipitates in good yields (93% and 95% for **S1** and **S2**, respectively). Similarly, the corresponding ligands **L1** and **L3**, and **L2** and **L3** were assembled with $\text{Cd}(\text{II})$ to produce complexes **S3** and **S4**.

The characterization of all ligands and complexes by NMR spectroscopy is shown in Figure 1 and Figures S2–S34. The ^1H NMR spectra of the ligands **L1**–**L3** and complexes **S1** and **S2** are shown in Figure 1. Owing to the modification on the 6-position of tpy, three sets of signals for pyridine corresponding to ligand **L3** were observed. For the two symmetrically modified guest ligands **L1** and **L2**, only two sets of signals for

pyridine were observed, indicative of the successful preparation of these designed ligands. ^1H NMR spectra of complexes **S1** and **S2** exhibited sharp and well-resolved proton signals, providing another strong indication of the formation of discrete assemblies. All of the signals for these complexes were accurately assigned by 2D correlation spectroscopy (COSY). The proton signals of 6-tpy in the ditopic ligands **L3** shifted upfield ($\Delta\delta \approx 2.52$ ppm for **S2**) due to the strong electron-shielding effect.^{37–41} Meanwhile, other protons showed downfield shifts, such as the signal of protons of d and d', due to the reduced electron density after coordination. In addition, the restrictive structure induced a split of NMR signals. For example, the signal of the tpy proton, such as proton C3, C4, C5, and C6 in **L1** or proton W3, W4, W5, and W6 in **L2**, splits into two sets of peaks, which indicates their rotation restriction caused by the altered chemical environment.

ESI-MS (electron ionization-mass spectrometry) and TWIM-MS (traveling-wave ion mobility-mass spectrometry) were utilized to further characterize the molecular composition and structural information. The ESI-MS of **S1** and **S2** shows one prominent set of peaks with charge states from 2+ to 4+ for **S1** and from 4+ to 7+ for **S2** (Figure 2a,c) due to the loss of a different number of PF_6^- . The experimental m/z values for each peak were perfectly consistent with the theoretical ones derived from the assembled composition of **S1** and **S2** with a molecular weight of 2694.6 and 5057.0 Da. The TWIM-MS experiments afforded additional structural evidence in which a series of consecutive charge signal distributions from complexes **S1** and **S2** with a narrow drift time were discernible (Figure 2b,d), indicating the formation of discrete structures without other isomers or structural conformers. ESI characterization of $\text{Cd}(\text{II})$ -based complexes **S3** and **S4** showed similar results. Through the above structural characterization, heteroleptic complexes **S1**–**S4** were evidenced to be the exclusive product rather than the homoleptic coordination

oligomers, indicating that such multicomponent assemblies exhibit high selectivity.

Although complexes **S1**–**S4** were characterized by NMR, ESI-MS, and TWIM-MS, detailed structural information about molecular configuration and the distance between the TPE core and binaphthyl motifs is still urgently needed. The single-crystal characterization has been proven to be the most powerful evidence to confirm the absolute configuration of molecules. All attempts to obtain single crystals of racemic complexes **S1**–**S4** were unsuccessful. Fortunately, the single crystal of (*S*)-**S2** suitable for X-ray diffraction was successfully obtained by slow vapor diffusion of isopropyl ether into an acetonitrile solution of enantiomerically pure (*S*)-**S2**. As expected, the X-ray crystallographic analysis showed that two binaphthyl-based ligands **L3** coordinate with the TPE-based ligand **L2** from the upper and lower directions, respectively, to form a heteroleptic supramolecular assembly (Figure 3). The

enantiomerically pure complex (*S*)-**S2** crystallized into the chiral space group *P*1. To accommodate the rigid ligand **L2** with a fixed arm length, the axially chiral binaphthyl core with an adaptable configuration adopted a specific twist, where the dihedral angle of two different naphthyl planes was $\sim 115.4^\circ$. The two methoxy groups on one binaphthyl ligand adopted the orientation outward the sterically crowded cavity of complex **S2**, and they are symmetrical, which are consistent with only one peak of the methoxy group observed in the ^1H NMR spectra. Two sets of C–H $\cdots\pi$ interactions were observed between the binaphthyl and TPE units. Most importantly, the distance between the phenyl of 6-position modification in **L3** and tpy group in **L2** is approximately 2.8 Å, which is close enough to effectively restrict the rotation of TPE, conforming to well-ordered and dense fluorophore arrangements.

Following structural confirmation of the complexes, the photophysical properties of all ligands and complexes were first examined in a dilute solution (Figure 4). Both TPE-based ligands (**L1** and **L2**) exhibited similar absorption bands in the range of 250–400 nm. The bands at approximately 280 and 325 nm correspond to the π – π^* transitions of the tpy groups and the conjugated TPE core, whereas the absorption between 350 and 420 nm is attributed to intramolecular charge transfer (ICT) from the electron-donating TPE moiety to the electron-accepting tpy units. The absorption spectrum of the binaphthyl-based ligand **L3** displayed broad bands spanning 250–350 nm, which are associated with electronic transitions of the tpy and binaphthyl units (Figure 4a).⁴² In photoluminescence (PL) measurements, **L3** showed emission bands with a maximum around 385 nm and a PLQY of 30.2% (Figure 4b). In contrast, TPE-based ligands **L1** and **L2** displayed very weak emission in solution. This weak fluorescence is due to unrestricted rotation of the TPE units, which promotes nonradiative decay and effectively quenches their emission.⁴³

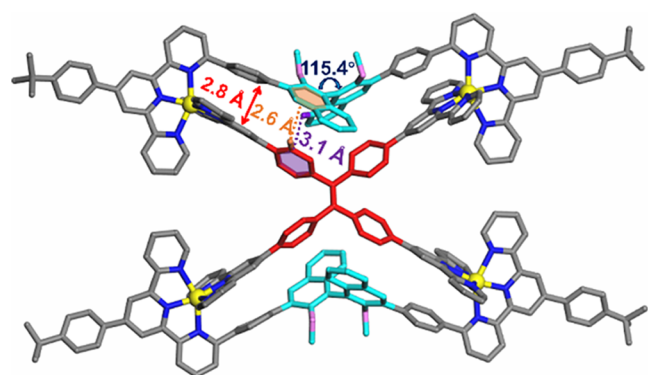


Figure 3. X-ray crystal structure of (*S*)-**S2**.

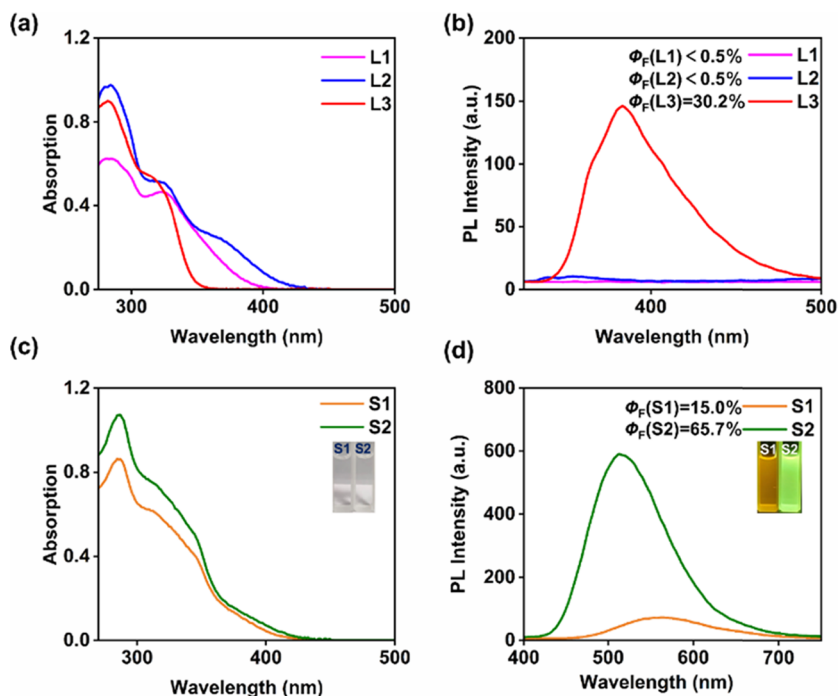


Figure 4. (a) Absorption and (b) fluorescent emission spectra of ligands **L1**, **L2**, and **L3** at the concentration of 10 μM in CHCl_3 . (c) Absorption and (d) fluorescent emission spectra of complexes **S1** and **S2** at the concentration of 10 μM in CH_3CN . Inset: photograph of complexes **S1** and **S2** in CH_3CN at the concentration of 10 μM upon daylight or excitation at 365 nm at 298 K.

After coordination with Zn(II) ions, the absorption peaks of S1 and S2 in CH₃CN exhibited a slight red shift compared to their respective ligands (Figure 4c). The sandwich-like complex S2 displayed an emission band at 530 nm with an enhanced Φ_F of 65.7% (Figure 4d), which was significantly greater than ligand L2 (0.5% for L2). It is noteworthy that this Φ_F is also higher than that of previously reported TPE-based decker structures, highlighting the effectiveness of the binaphthyl group in restricting the free rotation of TPE. The nonplanar structure of the binaphthyl group likely facilitates strong interactions with TPE, thereby effectively restricting its free rotation. While S1 exhibited an enhanced emission compared to ligand L1, Φ_F remained moderate (15.0% for S1) and was significantly lower than that of S2. This difference can be attributed to the fact that in S1, only two phenyl rings of the TPE core engage in close stacking with the binaphthyl groups, leaving the other two phenyl rings unrestricted. Consequently, the insufficient restriction fails to fully suppress nonradiative decay pathways, resulting in the lower Φ_F . Owing to its more pronounced fluorescence enhancement, complex S2 holds greater potential for application in turn-on fluorescence sensing of metal ions.

In addition, the time-resolved decayed lifetime (τ_s), radiative decay rate (k_r), and nonradiative decay rate (k_{nr}) of ligands and complexes were further investigated in CHCl₃ and CH₃CN, respectively (Figure 5, Figures S45 and S46, and Tables S2 and S3). The τ_s values of TPE-based ligands L1 and L2 in CHCl₃ were 1.25 and 1.09 ns, respectively. The k_{nr} values of ligands L1 and L2 were approximately calculated to be 7.99×10^8 and 9.17×10^8 s⁻¹, respectively. In contrast, the k_{nr} values of complexes S1–S4 were 2.81×10^8 , 4.06×10^7 , 2.95×10^8 , and 9.39×10^7 s⁻¹, respectively, which were considerably less than that of ligands. Even for S2 and S4, the k_{nr} values of them have decreased by as much as an order of magnitude. This result indicated that complexes S1–S4 might effectively suppress the energy loss via a nonradiative transition process by restricting the intramolecular rotation or vibration of TPE moieties. It is worth mentioning that the values of k_{nr}/k_r of S2 and S4 are smaller than those of S1 and S2, which might be attributed to a more restrictive chemical environment.

Based on the significant differences in fluorescence properties between the complexes and their corresponding ligands, we propose employing these ligands as turn-on fluorescence sensing materials. Initial selectivity screening was performed against 20 distinct metal ions, including Cd(II), Zn(II), Na(I), K(I), Ca(II), Ir(III), Tb(III), Eu(III), Pb(II), Os(II), Ag(I), Mg(II), Al(III), Cu(II), Ni(II), Mn(II), and Co(II). The tests were conducted by mixing each metal ion with two ligands (L2 and L3) at a precise stoichiometric ratio of L2/L3/M = 1:2:4. Notably, the ligands effectively distinguish Zn(II) and Cd(II) from other ions by eliciting an approximately 20-fold enhancement in fluorescence intensity, whereas most other metal ions induce negligible changes in emission (Figure 6a–c). Furthermore, a rapid fluorescence turn-on response was observed immediately after the addition of Zn(II) or Cd(II) to the ligand solution, indicating exceptionally high sensitivity. These results demonstrate that ligands L2 and L3 of complexes S2 and S4 can serve as highly selective turn-on fluorescent probes for the detection of Zn(II) and Cd(II).

Subsequent fluorescence titration experiments (Figure 6d–i) demonstrated that as Zn(II) or Cd(II) was gradually added into the mixed ligands (L2/L3 = 1:2) solution, the emission intensity at 515 nm increased significantly, especially within

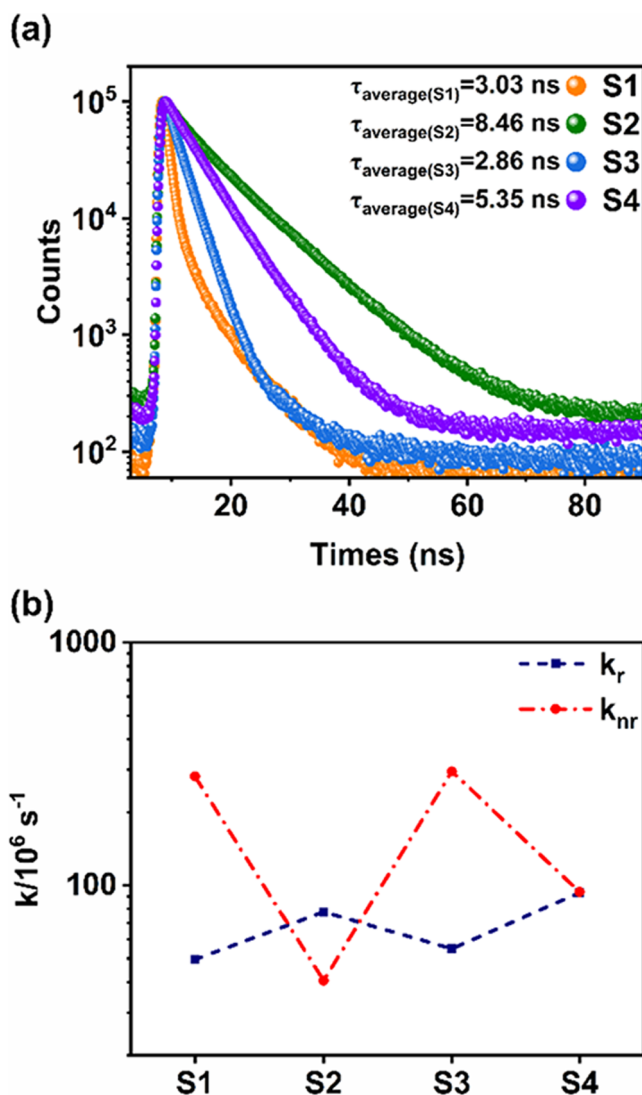


Figure 5. (a) Fluorescence decays for complexes S1, S2, S3, and S4 in CH₃CN at 298 K. (b) The radiative and nonradiative rate constants calculated from τ and Φ_F . k_r = radiative decay rate (Φ_F/τ_s); k_{nr} = nonradiative decay rate ($1/\tau_s - k_r$).

the concentration range of 0–30 μ M. Importantly, within this range, the fluorescence intensity exhibited a linear correlation with the concentration of Zn(II) or Cd(II), confirming the suitability of the ligands for quantitative detection. The calculated limits of detection (LODs) for Zn(II) and Cd(II) were approximately 1.7×10^{-7} and 1.8×10^{-7} M based on the linear calibration plots (Figure 6f,i). The LOD was determined using the formula $LOD = 3\sigma/k$, where σ represents the standard deviation of 20 measurements of blank samples and k denotes the slope of the calibration curve. By comparison, the ligand components L1 and L3 exhibited LODs of only 1.0×10^{-6} and 1.1×10^{-6} M for Zn(II) and Cd(II), respectively, indicating lower sensitivity than the L2 and L3 system (Figures S49–S51).

CONCLUSIONS

This study presents a turn-on fluorescence sensor for Zn(II) and Cd(II) based on a multicomponent supramolecular self-assembly. The TPE unit was incorporated into ligands L1 and L2 as the sensing core. Through heteroleptic assembly with

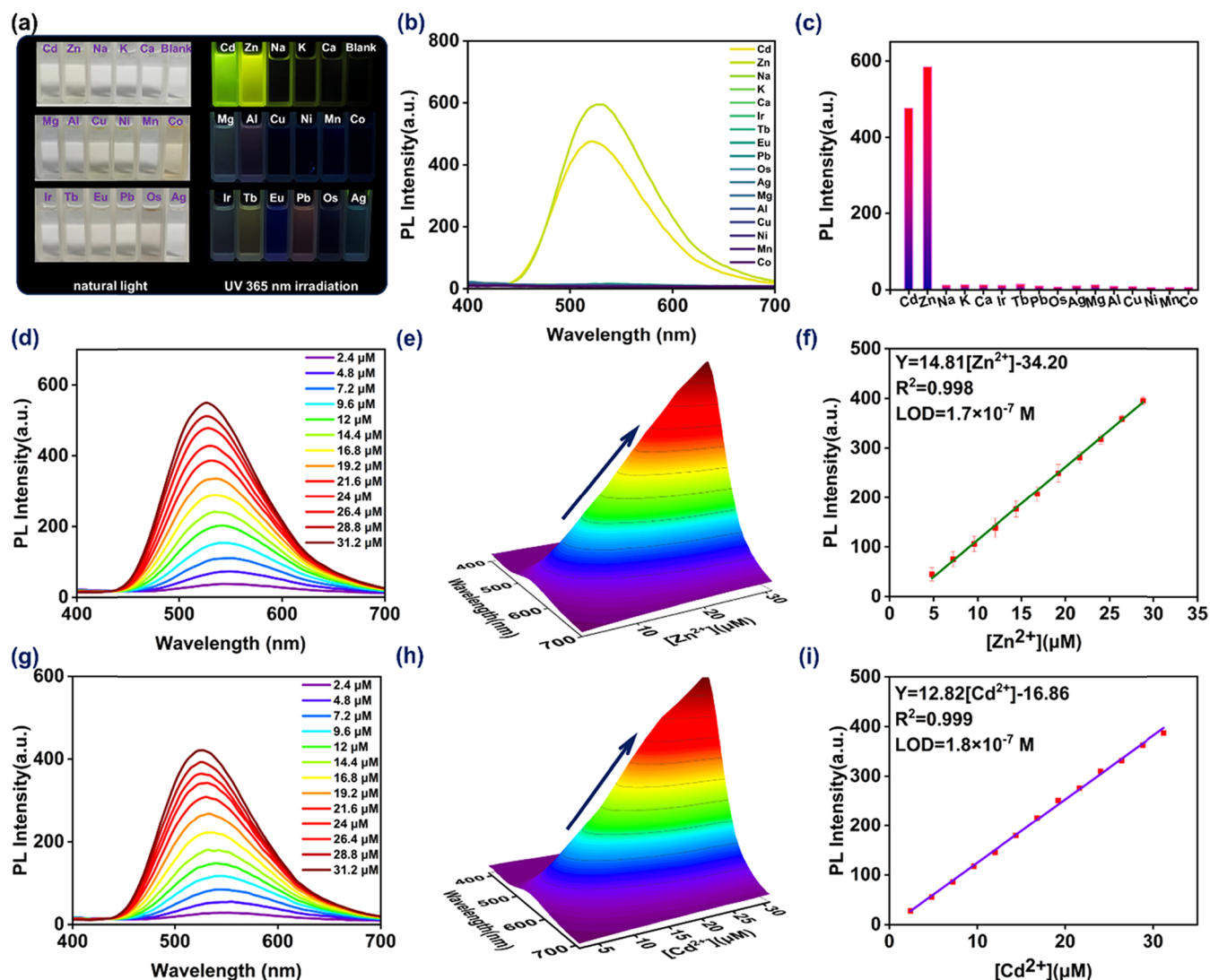


Figure 6. (a) Photographs of the metal ion test dripped in the ligand (L2 and L3) under daylight (left) and upon excitation at 365 nm (right). (b) Fluorescence emission spectra of ligands (L2 and L3) mixed with various metal ions. (c) PL intensity of the metal ion test dripped in ligands under daylight (left) and upon excitation at 365 nm. (d) Fluorescence emission spectra and (e) the 3D contour of ligands (L2 and L3) mixed with various concentrations of Zn(II) (0–30 μM). (f) The linear relationship between the fluorescence intensity of ligands (L2 and L3) and concentrations of Zn(II) in the range of 0–30 μM . (g) Fluorescence emission spectra and (h) the 3D contour of ligands (L2 and L3) mixed with various concentrations of Cd(II) (0–30 μM). (i) The linear relationship between the fluorescence intensity of ligands (L2 and L3) and concentrations of Cd(II) in the range of 0–30 μM .

binaphthyl ligand L3 and metal ions, supramolecular complexes S1–S4 with well-ordered and dense fluorophore arrangements were formed. The structural information on ligands and metallo-complexes was confirmed by NMR (^1H , ^{13}C , 2D COSY, and 2D DOSY), ESI-MS, TWIM-MS, and X-ray crystallography. The single crystal of enantiomerically pure complex (S)-S2 showed that coordination positions the binaphthyl units adjacent to the TPE, restricting intramolecular motion and enhancing fluorescence. Complexes S2 ($\Phi_F = 65.7\%$) and S4 ($\Phi_F = 49.8\%$) show higher efficiency due to complete restriction of all four TPE phenyl rings, while S1 (15.0%) and S3 (15.7%) with only two rings restricted exhibit moderate enhancement. Based on this spatial confinement strategy, the L2/L3-based systems function as effective sensors with LODs of 1.7×10^{-7} and 1.8×10^{-7} M for Zn(II) and Cd(II), respectively, outperforming the L1/L3 ligand components (1.0×10^{-6} and 1.1×10^{-6} M). This work not only provides an AIE-based assembly strategy for designing

selective fluorescent sensors through coordination-induced spatial restriction but also offers an effective strategy for constructing high-efficiency luminescent materials. By further tailoring ligand design and assembly pathways, multifunctional responsive sensing platforms may be developed, advancing potential applications in environmental monitoring and bioanalysis.

EXPERIMENTAL SECTION

Nuclear magnetic resonance (NMR) spectrum data were recorded at 300 K on Bruker 500 MHz and Qone AS 400 MHz nuclear magnetic resonance instruments using CDCl_3 and CD_3CN as the solvents. Electrospray ionization mass spectrometry (ESI-MS) spectra were collected on a Waters Synapt G2 mass spectrometer using solutions of 0.8 mg sample in 500 μL of acetonitrile for the complexes. MALDI-TOF-MS was performed on a Bruker AutoflexIII using trans-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenyli-dene] malononitrile (DCTB) as a matrix. The matrix was dissolved in CHCl_3 at 20 mg mL^{-1} , and organic compounds were dissolved in CHCl_3 at 10 mg

mL^{-1} . The sample was prepared by depositing $0.5 \mu\text{L}$ of the matrix on the wells of a 384-well ground-steel plate, allowing the spots to dry, and depositing $0.5 \mu\text{L}$ of the sample on a spot of dry matrix. The plate was inserted into the MALDI source after drying. The sample was conducted in linear mode. The data analysis was conducted with Bruker's FlexAnalysis software. Elemental analyses (carbon, hydrogen, nitrogen, sulfur) were performed on the elemental analyzer Elemental Vario EL cube. X-ray diffraction data for **S2** were collected using synchrotron radiation and a MAR325 CCD detector at the Shanghai Synchrotron Radiation BL17B Beamline. Indexing was performed by using APEX3. Data integration and reduction were performed using SAINTplus. Absorption correction was performed by a multiscan method implemented in SADABS. The space group was determined using XPREP implemented in APEX3.1. The structure was solved using SHELXT4 and was refined using SHELXL-2018 (full-matrix least-squares on F^2) through the OLEX2 interface program. Some of the disorder counterions and solvent molecules were refined with restraints. Ellipsoid plots were done with Platon. All ultraviolet–visible (UV–vis) spectra of solution samples were recorded on a PerkinElmer Lambda 365 spectrophotometer at room temperature. All emission spectra of solution samples were recorded on a Shimadzu RF-5301pc spectrophotometer. All PLQYs of solution samples and PLQYs were determined by an FLS980 spectrometer (Edinburgh Instruments Ltd., Livingston, United Kingdom). PLQY was measured via an integrating light sphere. Solutions were placed in 1 cm path length quartz cells.

Compound 2

The synthetic procedure was reported in ref 19. Ligands **L1** and **L2** were prepared according to ref 26.

Synthesis of Ligand L3. A mixture of compound **1** (0.3 mmol, 141 mg), compound **2** (0.72 mmol, 410 mg), and $\text{Pd}(\text{PPh}_3)_2\text{Cl}_2$ (0.03 mmol, 25 mg) in a 100 mL Schlenk flask was degassed three times. Then toluene (12 mL), 1 M Na_2CO_3 (6 mL), and *tert*-butyl alcohol (3 mL) were added under N_2 . The resultant mixture was kept at 85°C for 48 h. After cooling to room temperature, the solution was extracted by CHCl_3 three times, and the solvent was removed under reduced pressure. The crude product was purified by column chromatography on a silica gel ($\text{CHCl}_3/\text{EtOH} = 300/1$) to give ligand **L3** as a white solid (250 mg, 72%). ^1H NMR (500 MHz, CDCl_3 , 300 K) δ 9.02 (d, $J = 1.6$ Hz, 1H, H^{d}), 8.82–8.77 (m, 2H, H^{d} and H^{e}), 8.75 (d, $J = 8.0$ Hz, 1H, H^{b}), 8.69 (d, $J = 7.7$ Hz, 1H, H^{b}), 8.37 (d, $J = 8.0$ Hz, 2H, H^{b}), 8.12 (s, 1H, H^{e}), 8.05–7.98 (m, 4H, H^{c} , H^{d} and H^{b}), 7.94 (dd, $J = 9.7$, 5.9 Hz, 4H, H^{d} , H^{c} and H^{e}), 7.60 (d, $J = 8.1$ Hz, 2H, H^{f}), 7.48 (ddd, $J = 8.2$, 6.0, 1.9 Hz, 1H, H^{f}), 7.40 (t, $J = 6.2$ Hz, 1H, H^{f}), 7.34 (d, $J = 6.1$ Hz, 2H, H^{f} and H^{g}), 3.33 (s, 3H, H^{f}), 1.41 (s, 9H, H^{g}). ^{13}C NMR (125 MHz, CDCl_3 , 300 K) δ 156.72, 156.37, 156.35, 156.31, 156.06, 154.44, 152.57, 150.44, 149.42, 139.99, 138.55, 137.98, 137.20, 136.13, 134.86, 134.07, 131.19, 130.88, 130.05, 128.49, 127.79, 127.75, 127.38, 127.28, 126.77, 126.40, 126.29, 126.15, 125.45, 124.08, 121.72, 120.61, 119.99, 119.35, 119.21, 61.14, 35.08, 31.67. MALDI-TOF MS (m/z): calcd for $[\text{C}_{84}\text{H}_{68}\text{N}_6\text{O}_2 + \text{H}]^+$ 1193.23; found: 1193.23. Elemental analysis for ligand **L3** ($\text{C}_{84}\text{H}_{68}\text{N}_6\text{O}_2$): C 84.50% (calcd: 84.53%), H 5.84% (calcd: 5.74%), N 7.01% (calcd: 7.01%), S 0.00% (calcd: 0.00%).

Synthesis of Complex S1. To a solution of **L1** (8.6 mg, $10.8 \mu\text{mol}$) and **L3** (12.9 mg, $10.8 \mu\text{mol}$) in CHCl_3 (4 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (12.9 mg, $43.3 \mu\text{mol}$) in MeOH (8 mL) was added, and then the mixture was kept at 50°C for 12 h. After the mixture was cooled to room temperature, excess NH_4PF_6 (around 400 mg) was added to generate a white precipitate (27 mg, 93%). ^1H NMR (600 MHz, CD_3CN) δ 9.15 (d, $J = 1.4$ Hz, 1H, H^{d}), 8.94–8.90 (m, 2H, H^{d} and H^{c}), 8.89 (s, 1H, H^{CE}), 8.80 (d, $J = 8.0$ Hz, 1H, H^{C3}), 8.67 (d, $J = 8.2$ Hz, 1H, H^{C3}), 8.56 (d, $J = 8.2$ Hz, 1H, H^{f}), 8.50 (d, $J = 8.2$ Hz, 1H, H^{f}), 8.43–8.38 (m, 2H, H^{d} and H^{CE}), 8.38–8.31 (m, 2H, H^{C4} and H^{e}), 8.24–8.18 (m, 3H, H^{b} and H^{C4}), 8.05–8.00 (m, 1H, H^{f}), 8.00–7.98 (m, 1H, H^{C6}), 7.87–7.82 (m, 2H, H^{c}), 7.75 (t, $J = 7.5$ Hz, 1H, H^{d}), 7.66 (dd, $J = 10.3$, 6.5 Hz, 2H, H^{c} and H^{C4}), 7.58–7.53 (m, 1H, H^{C5}), 7.50–7.45 (m, 1H, H^{C5}), 7.33 (t, $J = 8.1$ Hz, 1H, H^{c}), 7.25–7.20 (m, 1H, H^{f}), 7.16–7.10 (m, 5H, H^{b} , H^{CA} and H^{CD}), 7.05

(s, 1H, H^{e}), 6.95 (dd, $J = 6.6$, 2.9 Hz, 4H, H^{CC} and H^{CB}), 6.73 (s, 1H, H^{f}), 6.35 (d, $J = 8.7$ Hz, 2H, H^{e} and H^{CG}), 2.82 (s, 3H, H^{f}), 1.49 (s, 9H, H^{g}). ^{13}C NMR (150 MHz, CD_3CN) δ 160.62, 155.94, 155.20, 155.07, 153.23, 151.09, 150.14, 149.77, 148.18, 148.17, 148.02, 147.91, 147.83, 147.32, 147.19, 145.68, 143.29, 142.87, 141.58, 141.45, 141.18, 140.65, 139.34, 138.94, 137.01, 133.23, 133.19, 133.05, 132.34, 131.62, 131.54, 130.55, 130.49, 128.79, 128.28, 127.93, 127.86, 127.80, 127.74, 127.49, 127.39, 127.35, 127.08, 127.05, 126.92, 126.74, 126.71, 125.87, 125.58, 125.47, 123.81, 123.05, 122.86, 122.73, 121.56, 121.41, 120.88, 120.31, 117.31, 60.09, 34.75, 30.44. ESI-MS (m/z): 1204.8 $[\text{M} - 2\text{PF}_6]^{2+}$ (calcd m/z : 1204.8), 755.2 $[\text{M} - 3\text{PF}_6]^{3+}$ (calcd m/z : 755.2), 530.2 $[\text{M} - 4\text{PF}_6]^{4+}$ (calcd m/z : 530.2).

Synthesis of Complex S2. To a solution of **L2** (5.6 mg, $4.4 \mu\text{mol}$) and **L3** (10.6 mg, $8.9 \mu\text{mol}$) in CHCl_3 (4 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (5.3 mg, $17.6 \mu\text{mol}$) in MeOH (8 mL) was added, and then the mixture was kept at 50°C for 12 h. After the mixture was cooled to room temperature, excess NH_4PF_6 (around 200 mg) was added to generate a white precipitate (21 mg, 95%). ^1H NMR (600 MHz, CD_3CN , 300 K) δ 9.15 (s, 1H, H^{d}), 8.91 (d, $J = 10.7$ Hz, 2H, H^{c} and H^{d}), 8.86 (s, 1H, H^{WE}), 8.80 (d, $J = 7.9$ Hz, 1H, H^{W3}), 8.66 (d, $J = 8.1$ Hz, 1H, H^{W3}), 8.52 (d, $J = 7.4$ Hz, 2H, H^{f} and H^{c}), 8.42–8.30 (m, 4H, H^{f} , H^{e} and H^{W4}), 8.20 (d, $J = 8.3$ Hz, 3H, H^{b} and H^{W4}), 7.97 (d, $J = 5.0$ Hz, 1H, H^{W6}), 7.90 (t, $J = 7.8$ Hz, 1H, H^{f}), 7.81 (dd, $J = 30.7$, 7.6 Hz, 3H, H^{c} and H^{f}), 7.65 (dd, $J = 16.6$, 6.3 Hz, 2H, H^{c} and H^{W6}), 7.54 (t, $J = 6.4$ Hz, 1H, H^{W3}), 7.46 (t, $J = 6.4$ Hz, 1H, H^{W5}), 7.35 (t, $J = 7.7$ Hz, 1H, H^{c}), 7.06–6.97 (m, 2H, H^{b} and H^{f}), 6.71 (s, 2H, H^{e} and H^{f}), 6.27 (d, $J = 8.7$ Hz, 1H, H^{e}), 2.81 (s, 3H, H^{e}), 1.48 (s, 10H, H^{g}). ^{13}C NMR (150 MHz, CD_3CN , 300 K) δ 160.60, 155.96, 155.12, 154.85, 153.20, 151.10, 150.06, 149.73, 148.24, 148.16, 148.01, 147.94, 147.87, 147.26, 147.12, 144.82, 141.58, 141.48, 141.19, 141.10, 140.51, 139.58, 139.33, 136.98, 133.43, 133.17, 133.02, 132.26, 131.52, 130.46, 128.74, 128.29, 127.92, 127.81, 127.79, 127.76, 127.24, 127.04, 126.71, 125.85, 125.62, 125.45, 123.88, 123.11, 122.81, 122.74, 121.61, 121.44, 120.77, 120.16, 60.08, 34.75, 30.42. ESI-MS (m/z): 1122.1 $[\text{M} - 4\text{PF}_6]^{4+}$ (calcd m/z : 1122.1), 868.6 $[\text{M} - 5\text{PF}_6]^{5+}$ (calcd m/z : 868.6), 699.9 $[\text{M} - 6\text{PF}_6]^{6+}$ (calcd m/z : 699.9), 579.2 $[\text{M} - 7\text{PF}_6]^{7+}$ (calcd m/z : 579.2).

Synthesis of Complex S3. To a solution of **L1** (3.6 mg, $4.5 \mu\text{mol}$) and **L3** (5.4 mg, $4.5 \mu\text{mol}$) in CHCl_3 (3.0 mL), a solution of $\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ (2.8 mg, $9 \mu\text{mol}$) in MeOH (6 mL) was added, and then the mixture was kept in a 50°C for 12 h. After the mixture was cooled to room temperature, excess NH_4PF_6 (around 190 mg) was added to generate a white precipitate (11 mg, 91%). ^1H NMR (600 MHz, CD_3CN , 300 K) δ 9.09 (d, $J = 1.4$ Hz, 1H, H^{d}), 8.90 (dd, $J = 4.7$, 3.3 Hz, 2H, H^{d} and H^{c}), 8.83 (s, 1H, H^{CE}), 8.77 (dd, $J = 11.1$, 8.2 Hz, 2H, H^{C3} and H^{C3}), 8.60 (d, $J = 8.3$ Hz, 1H, H^{f}), 8.48–8.39 (m, 4H, H^{f} , H^{c} , H^{CE} and H^{C4}), 8.18 (dt, $J = 7.7$, 2.8 Hz, 4H, H^{b} , H^{C4} and H^{C6}), 8.12 (s, 1H, H^{e}), 8.07–8.02 (m, 1H, H^{f}), 7.91–7.87 (m, 1H, H^{C6}), 7.85–7.79 (m, 3H, H^{c} and H^{c}), 7.74–7.69 (m, 1H, H^{f}), 7.68–7.59 (m, 3H, H^{b} , H^{CS} and H^{CE}), 7.49–7.44 (m, 1H, H^{CS}), 7.28–7.21 (m, 2H, H^{f} and H^{b}), 7.16–7.06 (m, 6H, H^{a} , H^{CD} and H^{CC}), 7.01–6.92 (m, 4H, H^{CB} and H^{e}), 6.38 (d, $J = 7.9$ Hz, 2H, H^{f}), 6.32 (d, $J = 8.6$ Hz, 1H, H^{e}), 2.68 (d, $J = 1.2$ Hz, 3H, H^{f}), 1.47 (s, 9H, H^{g}). ^{13}C NMR (150 MHz, CD_3CN) δ 160.31, 155.20, 154.79, 154.16, 153.00, 151.15, 150.57, 150.26, 149.52, 149.35, 149.16, 148.80, 148.57, 148.23, 148.07, 147.78, 145.98, 142.67, 141.82, 141.58, 141.00, 140.71, 139.35, 138.94, 138.52, 133.28, 133.25, 133.18, 132.63, 131.71, 131.34, 130.69, 129.85, 128.71, 128.67, 127.84, 127.53, 127.38, 127.03, 126.93, 126.85, 126.65, 125.78, 125.60, 125.49, 124.15, 123.33, 123.11, 123.06, 121.93, 121.76, 121.24, 120.63, 117.31, 59.93, 34.71, 30.44. ESI-MS (m/z): 786.5 $[\text{M} - 3\text{PF}_6]^{3+}$ (calcd m/z : 786.5), 553.7 $[\text{M} - 4\text{PF}_6]^{4+}$ (calcd m/z : 553.7).

Synthesis of Complex S4. To a solution of **L2** (4.3 mg, $3.4 \mu\text{mol}$) and **L3** (8.2 mg, $6.8 \mu\text{mol}$) in CHCl_3 (3 mL), a solution of $\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ (4.2 mg, $13.7 \mu\text{mol}$) in MeOH (6 mL) was added, and then the mixture was kept at 50°C for 12 h. After cooling to room temperature, excess NH_4PF_6 (around 273 mg) was added to generate a white precipitate (16.9 mg, 95%). ^1H NMR (600 MHz,

CD₃CN, 300 K) δ 9.08 (s, 1H, H^d), 8.94–8.86 (m, 2H, H^{3'} and H^d), 8.83–8.72 (m, 3H, H^{WE}, H^{W3'} and H^{W3}), 8.56 (d, J = 8.4 Hz, 1H, H^b), 8.52–8.36 (m, 4H, H³, H^c, H^{W4} and H^{WE}), 8.21–8.10 (m, 5H, H^b, H^{W4'}, H^{W6} and H⁸), 7.98 (t, J = 8.0 Hz, 1H, Hⁱ), 7.88 (d, J = 4.9 Hz, 1H, H^{W6'}), 7.81 (d, J = 7.9 Hz, 3H, H^c and H^{5'}), 7.75 (t, J = 7.5 Hz, 1H, Hⁱ), 7.63 (d, J = 6.7 Hz, 1H, H^{W5}), 7.56 (d, J = 4.8 Hz, 1H, Hⁱ), 7.49–7.42 (m, 1H, H^{W5'}), 7.26 (t, J = 7.8 Hz, 1H, H^{W5}), 7.11 (dd, J = 24.4, 5.8 Hz, 4H, H^{WA}, Hⁱ, H^e, H^f and H^{WB}), 6.27 (d, J = 8.7 Hz, 1H, H⁶), 2.68 (s, 3H, H²), 1.47 (s, 9H, H³). ¹³C NMR (150 MHz, CD₃CN, 300 K) δ 160.28, 155.20, 154.80, 153.98, 152.99, 151.16, 150.54, 150.21, 149.51, 149.38, 149.15, 148.79, 148.44, 148.24, 148.01, 147.73, 144.90, 141.82, 141.59, 141.00, 140.63, 140.44, 139.36, 138.47, 133.63, 133.26, 133.23, 132.56, 131.54, 131.35, 129.85, 128.66, 127.82, 127.54, 127.39, 127.01, 126.83, 126.64, 125.77, 125.65, 125.49, 124.21, 123.31, 123.14, 123.06, 121.95, 121.75, 121.23, 120.53, 59.92, 34.71, 30.43. ESI-MS (m/z): 1169.1 [M – 4PF₆]⁴⁺ (calcd m/z : 1169.1), 906.5 [M – 5PF₆]⁵⁺ (calcd m/z : 906.5), 731.2 [M – 6PF₆]⁶⁺ (calcd m/z : 731.2), 606.2 [M – 7PF₆]⁷⁺ (calcd m/z : 606.2).

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.6c00536>.

Experimental details, NMR and HR-MS data, X-ray diffraction data, and absorption and fluorescent emission spectra information (PDF)

A video animation illustrating the crystal structure of (S)-S2 (MP4)

Accession Codes

Deposition Number 2538777 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

■ AUTHOR INFORMATION

Corresponding Authors

Ningxu Han – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, China; Email: hannx25@jlu.edu.cn

Ming Wang – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, China; orcid.org/0000-0002-5332-0804; Email: mingwang358@jlu.edu.cn

Authors

Shaozhi Wang – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, China

Manman Dai – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, China

Ning Zhang – Jilin Province Product Quality Supervision and Inspection Institute, Changchun, Jilin 130103, China

Ziteng Guo – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, China

Desheng Li – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, China

Jingsong Yuan – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, China

Yibin Sun – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, China; orcid.org/0000-0003-0450-5391

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.inorgchem.6c00536>

Author Contributions

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Notes

The authors declare no competing financial interest.

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